

# Chapter 1

Connor Mooneyhan

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**Exercise 1.** A nonempty family of sets  $\mathcal{R} \in \mathcal{P}(X)$  is called a **ring** if it is closed under finite unions and differences (i.e., if  $E_1, \dots, E_n \in \mathcal{R}$ , then  $\cup_1^n E_j \in \mathcal{R}$ , and if  $E, F \in \mathcal{R}$ , then  $E \setminus F \in \mathcal{R}$ ). A ring that is closed under countable unions is called a  **$\sigma$ -ring**.

- a. Rings (resp.  $\sigma$ -rings) are closed under finite (resp. countable) intersections.
- b. If  $\mathcal{R}$  is a ring (resp.  $\sigma$ -ring), then  $\mathcal{R}$  is an algebra (resp.  $\sigma$ -algebra) iff  $X \in \mathcal{R}$ .
- c. If  $\mathcal{R}$  is a  $\sigma$ -ring, then  $\{E \subset X : E \in \mathcal{R} \text{ or } E^c \in \mathcal{R}\}$  is a  $\sigma$ -algebra.
- d. If  $\mathcal{R}$  is a  $\sigma$ -ring, then  $\{E \subset X : E \cap F \in \mathcal{R} \text{ for all } F \in \mathcal{R}\}$  is a  $\sigma$ -algebra.

*Proof.*

- a. If  $\mathcal{R}$  is a  $\sigma$ -ring and  $\{E_n\}_{n \in \mathbb{N}} \subset \mathcal{R}$ , then

$$\bigcap_{n \in \mathbb{N}} E_n = \left( \bigcup_{n \in \mathbb{N}} E_n \right) \setminus \left( \bigcup \{E_a \setminus E_b : a, b \in \mathbb{N}\} \right) \quad (1)$$

Considering the left and right components of the difference in the right hand side of (1), the left is clearly a countable union of members of  $\mathcal{R}$ , and the right can be seen to be a countable union by identifying each  $E_a \setminus E_b$  with  $\frac{a}{b}$ . Thus the right hand side of (1) is a difference of countable unions of members of  $\mathcal{R}$ , so it is in  $\mathcal{R}$ .

If  $\mathcal{R}$  is a ring, the argument above applies by making  $\{E_n\}$  a finite set.

- b. In either case, if  $\mathcal{R}$  is a ring (resp.  $\sigma$ -ring) and an algebra (resp.  $\sigma$ -algebra), then given  $E \in \mathcal{R}$ ,  $X = E \cup E^c \in \mathcal{R}$ .

Conversely, if  $\mathcal{R}$  is a ring (resp.  $\sigma$ -ring) and  $X \in \mathcal{R}$ , then given  $E \in \mathcal{R}$ ,  $E^c = X \setminus E \in \mathcal{R}$ , so  $\mathcal{R}$  is closed under complements.  $\mathcal{R}$  is already closed under finite (resp. countable) unions, so  $\mathcal{R}$  is an algebra.

- c. Let  $A = \{E \subset X : E \in \mathcal{R} \text{ or } E^c \in \mathcal{R}\}$ . Given  $E \in A$ , by definition  $E^c \in A$  since  $(E^c)^c = E \in A$ .

Let  $\{E_n\}_{n \in \mathbb{N}} \subset A$ , and express this set as the union of two sets, namely

$$\{E_n\}_{n \in \mathbb{N}} = \{F_j\}_{j \in J} \cup \{G_k\}_{k \in K}$$

where  $\{F_j\}_{j \in J} \subset A$  is the collection of all sets in  $A$  who are members of  $\mathcal{R}$  and  $\{G_k\}_{k \in K} \subset A$  is the collection of sets in  $A$  who are not themselves in  $\mathcal{R}$ , but whose complements are. Then

$$\begin{aligned} \bigcup_{n \in \mathbb{N}} E_n &= \bigcup_{j \in J} F_j \cup \bigcup_{k \in K} G_k \\ &= \bigcup_{j \in J} F_j \cup \left( \bigcap_{k \in K} G_k^c \right)^c \end{aligned} \quad (1)$$

Since each  $F_j$  is in  $\mathcal{R}$ ,  $\bigcup_{j \in J} F_j \in \mathcal{R} \subset A$ . Also, we know from (a) that  $\sigma$ -rings are closed under countable intersections, so since  $G_k^c \in \mathcal{R}$  for each  $G_k$ , we have  $\bigcap_{k \in K} G_k^c \in \mathcal{R}$ . Its complement is also in  $\mathcal{R}$  since  $\mathcal{R}$  is closed under complementation. Thus (2) is a union of two members of  $\mathcal{R}$ , and is therefore in  $\mathcal{R} \subset A$ .

- d. Let  $A = \{E \subset X : E \cap F \in \mathcal{R} \text{ for all } F \in \mathcal{R}\}$ . Note that  $\mathcal{R} \subset A$  since  $\mathcal{R}$  is closed under finite intersections. Given any  $E, F \in \mathcal{R}$ , we have

$$E^c \cap F = (X \setminus E) \cap F = F \setminus E$$

which is a difference in  $\mathcal{R}$  and is thus in  $\mathcal{R} \subset A$ .

Given  $\{E_j\}_{j \in \mathbb{N}} \subset A$  and arbitrary  $F \in A$ ,

$$\left( \bigcup_{j \in \mathbb{N}} E_j \right) \cap F = \bigcup_{j \in \mathbb{N}} (E_j \cap F),$$

which is a countable union of elements in  $\mathcal{R}$ . Thus since the choice of  $F$  was arbitrary,  $\bigcup \{E_j\} \in A$ .

□

**Exercise 2.** Complete the proof of Proposition 1.2.

*Proof.* The construction of open intervals from half-open intervals and rays is standard. □

**Exercise 3.** Let  $\mathcal{M}$  be an infinite  $\sigma$ -algebra.

- a.  $\mathcal{M}$  contains an infinite sequence of disjoint nonempty sets.
- b.  $\text{card}(\mathcal{M}) \geq c$ .

*Proof.*

- a. Choose a sequence  $\{E_n\}_{n \in \mathbb{N}}$  of nonempty sets in  $\mathcal{M}$  (this requires the axiom of choice). We then derive the sequence  $\{F_n\}$  defined recursively by  $F_1 = E_1$  and  $F_j = E_j \setminus F_{j-1}$  when  $j \geq 2$ . These are disjoint by construction and are all in  $\mathcal{M}$  since  $\mathcal{M}$  is closed under differences.
- b. Choose a sequence  $\{E_j\}_1^\infty$  of disjoint nonempty sets in  $\mathcal{M}$  (allowable by part (a) of this exercise). Define  $f : \mathcal{P}(\mathbb{N}) \rightarrow \mathcal{M}$  by  $f : S \mapsto \bigcup_{k \in S} E_k$ . This is well-defined since  $f(S)$  is a countable union of members of  $\mathcal{M}$ , so  $f(S) \in \mathcal{M}$ . Also,  $f$  is an injection since all the  $E_j$  are disjoint. This brings us to the conclusion that since  $\text{im } f \subset \mathcal{M}$ ,

$$\text{card } \mathcal{M} \geq \text{card}(\text{im } f) = \text{card}(\mathcal{P}(\mathbb{N})) = c.$$

□

**Exercise 4.** An algebra  $\mathcal{A}$  is a  $\sigma$ -algebra iff  $\mathcal{A}$  is closed under countable increasing unions (i.e., if  $\{E_j\}_1^\infty \subset \mathcal{A}$  and  $E_1 \subset E_2 \subset \dots$ , then  $\bigcup_1^\infty E_j \in \mathcal{A}$ ).

*Proof.* If  $\mathcal{A}$  is a  $\sigma$ -algebra, then  $\mathcal{A}$  is closed under countable unions in general, so it is certainly closed under countable increasing unions in particular. On the other hand, if  $\mathcal{A}$  is an algebra which is closed under countable increasing unions, then take any  $\{E_j\}_1^\infty \subset \mathcal{A}$ . Consider that

$$\bigcup_{j=1}^\infty E_j = \bigcup_{j=1}^\infty \bigcup_{k=1}^j E_k. \quad (1)$$

Clearly, for any  $j \in \mathbb{N}$ ,  $\left(\bigcup_{k=1}^j E_k\right) \subset \left(\bigcup_{k=1}^{j+1} E_k\right)$ , and each  $\bigcup_{k=1}^j E_k$  is a finite union of elements of the algebra  $\mathcal{A}$ , and thus is in  $\mathcal{A}$ . Then the right hand side of (1) is a countable increasing union of members of  $\mathcal{A}$ . Thus (1)  $\in \mathcal{A}$ . □

**Exercise 5.** If  $\mathcal{M}$  is the  $\sigma$ -algebra generated by  $\mathcal{E}$ , then  $\mathcal{M}$  is the union of the  $\sigma$ -algebras generated by  $\mathcal{F}$  as  $\mathcal{F}$  ranges over all countable subsets of  $\mathcal{E}$ .

*Proof.* Let  $F$  be the set of all countable subsets of  $\mathcal{E}$ . By Lemma 1.1, since  $\mathcal{F} \subset \mathcal{E} \subset \mathcal{M}(\mathcal{E})$  for each  $\mathcal{F} \in F$ ,  $\mathcal{M}(\mathcal{F}) \subset \mathcal{M}(\mathcal{E})$ . Their union, then, is also a subset of  $\mathcal{E}$ .

We need to show, then, that  $\mathcal{M}(\mathcal{E}) \subset \bigcup_{\mathcal{F} \in F} \mathcal{M}(\mathcal{F})$ . If we can show that  $\bigcup_{\mathcal{F} \in F} \mathcal{M}(\mathcal{F}) = \mathcal{M}(\bigcup_{\mathcal{F} \in F} \mathcal{F})$ , then we will be done because  $\mathcal{E} \subset \bigcup F$ , so  $\mathcal{M}(\mathcal{E}) \subset \mathcal{M}(\bigcup F)$ . That is where we now turn our focus.

Clearly  $\mathcal{M}(\bigcup F) \subset \bigcup_{\mathcal{F} \in F} \mathcal{M}(\mathcal{F})$  since  $\bigcup F \subset \bigcup_{\mathcal{F} \in F} \mathcal{M}(\mathcal{F})$ , so we need to prove the other direction. Given  $E \in \bigcup \mathcal{M}(F)$ , fix some  $\mathcal{F} \in F$  for which  $E \in \mathcal{M}(\mathcal{F})$ . Then since  $\mathcal{F} \subset \bigcup F$ ,  $\mathcal{M}(\mathcal{F}) \subset \mathcal{M}(\bigcup F)$ , so  $E \in \mathcal{M}(\bigcup F)$ . As observed above, this completes our proof. □

**Exercise 6.** Complete the proof of Theorem 1.9.

*Proof.* First we verify that  $\bar{\mu}$  is a complete measure on  $\bar{\mathcal{M}}$ . Since we can write  $\emptyset = \emptyset \cup \emptyset$  where  $\emptyset \in \mathcal{M}, \mathcal{N}$ , we have  $\bar{\mu} = \mu(\emptyset) = 0$ . Given a disjoint collection  $\{E_j \cup F_j\}_{j \in \mathbb{N}} \subset \bar{\mathcal{M}}$  where each  $E_j \in \mathcal{M}$  and each  $F_j \subset N_j \in \mathcal{N}$ , we have

$$\begin{aligned} \bar{\mu} \left( \bigcup_1^\infty (E_j \cup F_j) \right) &= \bar{\mu} \left( \bigcup_1^\infty E_j \cup \bigcup_1^\infty F_j \right) \\ &= \mu \left( \bigcup_1^\infty E_j \right) \\ &= \sum_1^\infty \mu(E_j) \\ &= \sum_1^\infty \bar{\mu}(E_j \cup F_j) \end{aligned}$$

where the second equality holds because  $\bigcup_1^\infty E_j \in \mathcal{M}$  and  $\bigcup_1^\infty F_j \in \mathcal{N}$  since both  $\mathcal{M}$  and  $\mathcal{N}$  are closed under countable unions. Of course, since  $\bar{\mu}$  is a measure and its domain is complete,  $\bar{\mu}$  is a complete measure.

Now we turn our attention to the uniqueness of  $\bar{\mu}$  as an extension of  $\mu$  over  $\bar{\mathcal{M}}$ . Assume that  $f$  is a measure over  $\bar{\mathcal{M}}$  which extends  $\mu$ . Then given  $M \cup F \in \bar{\mathcal{M}}$  where  $M \in \mathcal{M}$  and  $F \subset N \in \mathcal{N}$ , by monotonicity we have that  $f(F) \leq f(N) = 0$ , so  $f(F) = 0$ . We have

$$f(M) \leq f(M \cup F) \leq f(M) + f(F) = f(M).$$

Thus  $f(M \cup F) = f(M) = \mu(M) = \bar{\mu}(M \cup F)$ . □

**Exercise 7.** If  $\mu_1, \dots, \mu_n$  are measures on  $(X, \mathcal{M})$  and  $a_1, \dots, a_n \in [0, \infty)$ , then  $\sum_1^n a_j \mu_j$  is a measure on  $(X, \mathcal{M})$ .

*Proof.* Clearly  $\sum_1^n a_j \mu_j(\emptyset) = 0$ . Given a disjoint collection  $\{E_j\}_{j \in \mathbb{N}} \subset \mathcal{M}$ , we have

$$\begin{aligned} \sum_{k=1}^n \left( a_k \mu_k \left( \bigcup_{j=1}^\infty E_j \right) \right) &= \sum_{k=1}^n \left( a_k \sum_{j=1}^\infty \mu_k(E_j) \right) \\ &= \sum_{k=1}^n \sum_{j=1}^\infty a_k \mu_k(E_j) \\ &= \sum_{j=1}^\infty \sum_{k=1}^n a_k \mu_k(E_j). \end{aligned}$$

□

**Exercise 8.** If  $(X, \mathcal{M}, \mu)$  is a measure space and  $\{E_j\}_1^\infty \subset \mathcal{M}$ , then  $\mu(\liminf E_j) \leq \liminf \mu(E_j)$ . Also,  $\mu(\limsup E_j) \geq \limsup \mu(E_j)$  provided that  $\mu(\bigcup_1^\infty E_j) < \infty$ .

*Proof.* Observe that

$$\begin{aligned}
\mu(\liminf E_j) &= \mu\left(\bigcup_{k=1}^{\infty} \bigcap_{j=k}^{\infty} E_j\right) \\
&= \lim_{k \rightarrow \infty} \mu\left(\bigcap_{j=k}^{\infty} E_j\right) \\
&\leq \lim_{k \rightarrow \infty} \mu(E_k) \\
&\dots \text{ UNFINISHED} \\
&= \sup_{k \geq 1} \inf_{j \geq k} \mu(E_j) \\
&= \liminf \mu(E_j)
\end{aligned}$$

□

**Exercise 9.** If  $(X, \mathcal{M}, \mu)$  is a measure space and  $E, F \in \mathcal{M}$ , then  $\mu(E) + \mu(F) = \mu(E \cup F) + \mu(E \cap F)$ .

*Proof.*

$$\begin{aligned}
\mu(E \cup F) &= \mu((E \setminus F) \cup (E \cap F) \cup (F \setminus E)) \\
&= \mu(E \setminus F) + \mu(E \cap F) + \mu(F \setminus E) \\
&= \mu(E \setminus F) + \mu(E \cap F) - \mu(E \cap F) + \mu(E \cap F) + \mu(F \setminus E) \\
&= \mu((E \setminus F) \cup (E \cap F)) - \mu(E \cap F) + \mu((E \cap F) \cup (F \setminus E)) \\
&= \mu(E) - \mu(E \cap F) + \mu(F).
\end{aligned}$$

Thus,  $\mu(E \cup F) + \mu(E \cap F) = \mu(E) + \mu(F)$  as desired.

□

**Exercise 10.** Given a measure space  $(X, \mathcal{M}, \mu)$  and  $E \in \mathcal{M}$ , define  $\mu_E(A) = \mu(A \cap E)$  for  $A \in \mathcal{M}$ . Then  $\mu_E$  is a measure.

*Proof.* First off,  $\mu_E(\emptyset) = \mu(\emptyset \cap E) = \mu(\emptyset) = 0$ . Now suppose that we have some

$\{E_j\}_{j \in \mathbb{N}} \subset \mathcal{M}$  a disjoint collection. Then

$$\begin{aligned}
\mu_E \left( \bigcup_{j=1}^{\infty} E_j \right) &= \mu \left( \left( \bigcup_{j=1}^{\infty} E_j \right) \cap E \right) \\
&= \mu \left( \bigcup_{j=1}^{\infty} E_j \cap E \right) \\
&= \sum_{j=1}^{\infty} \mu(E_j \cap E) \\
&= \sum_{j=1}^{\infty} \mu_E(E_j).
\end{aligned}$$

□

**Exercise 11.** A finitely additive measure  $\mu$  is a measure iff it is continuous from below as in Theorem 1.8c. If  $\mu(X) < \infty$ ,  $\mu$  is a measure iff it is continuous from above as in Theorem 1.8d.

*Proof.* For both assertions, if  $\mu$  is a measure, then clearly it is continuous from above and below so we restrict ourselves to proving the other direction in either case. Also,  $\mu(\emptyset) = 0$  since  $\mu$  is a finitely additive measure in both cases, so we will only prove countable additivity.

Given that  $\mu$  is a finitely additive measure and it is continuous from below, consider a disjoint collection  $\{E_j\}_{j \in \mathbb{N}} \subset \mathcal{M}$ . Consider the sequence defined by  $F_j = \bigcup_{k=1}^j E_k$ , which is clearly an ascending chain. Then

$$\begin{aligned}
\mu \left( \bigcup_{j=1}^{\infty} E_j \right) &= \mu \left( \bigcup_{j=1}^{\infty} F_j \right) \\
&= \lim_{j \rightarrow \infty} \mu(F_j) \\
&= \lim_{j \rightarrow \infty} \mu \left( \bigcup_{k=1}^j E_k \right) \\
&= \lim_{j \rightarrow \infty} \sum_{k=1}^j \mu(E_k) \\
&= \sum_{j=1}^{\infty} \mu(E_j).
\end{aligned}$$

Similarly, assuming that  $\mu(X) < \infty$  and  $\mu$  is continuous from above, define

$\{F_j\}$  as above. Observe that  $\{F_j^C\}$  is a descending chain, so

$$\begin{aligned}
\mu\left(\bigcup_{j=1}^{\infty} E_j\right) &= \mu\left(\bigcup_{j=1}^{\infty} F_j\right) \\
&= \mu\left(\left(\bigcap_{j=1}^{\infty} F_j^C\right)^C\right) \\
&= \mu\left(X \setminus \left(\bigcap_{j=1}^{\infty} F_j^C\right)\right) \\
&= \mu(X) - \mu\left(\bigcap_{j=1}^{\infty} F_j^C\right) \\
&= \mu(X) - \lim_{j \rightarrow \infty} \mu(F_j^C) \\
&= \mu(X) - \lim_{j \rightarrow \infty} \mu(X \setminus F_j) \\
&= \mu(X) - \lim_{j \rightarrow \infty} (\mu(X) - \mu(F_j)) \\
&= \mu(X) - \lim_{j \rightarrow \infty} \mu(X) + \lim_{j \rightarrow \infty} \mu(F_j) \\
&= \lim_{j \rightarrow \infty} \mu(F_j) \\
&= \sum_{j=1}^{\infty} \mu(E_j).
\end{aligned}$$

□

**Exercise 12.** Let  $(X, \mathcal{M}, \mu)$  be a finite measure space.

- a. If  $E, F \in \mathcal{M}$  and  $\mu(E \triangle F) = 0$ , then  $\mu(E) = \mu(F)$ .
- b. Say that  $E \sim F$  if  $\mu(E \triangle F) = 0$ ; then  $\sim$  is an equivalence relation on  $\mathcal{M}$ .
- c. For  $E, F \in \mathcal{M}$ , define  $\rho(E, F) = \mu(E \triangle F)$ . Then  $\rho(E, G) \leq \rho(E, F) + \rho(F, G)$ , and hence  $\rho$  defines a metric on the space  $\mathcal{M}/\sim$  of equivalence classes.

*Proof.*

- a. Observe that

$$0 = \mu(E \triangle F) = \mu(E \cup F) - \mu(E \cap F),$$

so

$$\begin{aligned}
\mu(E \cap F) &= \mu(E \cup F) \\
&= \mu((E \setminus F) \cup F) \\
&= \mu(E \setminus F) + \mu(F) \\
&= \mu(F)
\end{aligned}$$

since  $E \setminus F \subset E \triangle F$ , and thus  $\mu(E \setminus F) = 0$  by monotonicity. Swapping  $E$  and  $F$  in the above calculation reveals that  $\mu(E \cap F) = \mu(E)$  as well, so  $\mu(F) = \mu(E)$ .

- b.** Surely given  $E \in \mathcal{M}$ ,  $\mu(E \triangle E) = \mu(\emptyset) = 0$ , so  $E \sim E$ . Given also  $F \in \mathcal{M}$  such that  $E \sim F$ , then since  $E \triangle F = F \triangle E$ ,  $\mu(F \triangle E) = 0$  so that  $F \sim E$ . Finally, if we have  $G \in \mathcal{M}$  such that  $F \sim G$ , then notice that

$$E \triangle G \subset (E \triangle F) \cup (F \triangle G).$$

The easiest way to see this is with a venn diagram (shoot me), but it can also be shown with some slightly painful set theory. From this, we use monotonicity and subadditivity (respectively) to get

$$\begin{aligned}
\mu(E \triangle G) &\leq \mu((E \triangle F) \cup (F \triangle G)) \\
&\leq \mu(E \triangle F) + \mu(F \triangle G) \\
&= 0,
\end{aligned}$$

so  $\mu(E \triangle G) = 0$ .

- c.** By the definition of  $\sim$ ,  $\rho(E, F) = 0$  if and only if  $E \sim F$ . Also,  $\rho(E, F) = E \triangle F = F \triangle E = \rho(F, E)$ , so the problem statement is justified in that we need only prove the triangle inequality.

The following is similar to the end of **b**, but is reproduced here without the specific choices of  $E, F, G$  used in **b** and adapted for the problem at hand. Given *any*  $E, F, G \in \mathcal{M}$ , we have

$$\begin{aligned}
\rho(E, G) &= \mu(E \triangle G) \\
&\leq \mu((E \triangle F) \cup (F \triangle G)) \\
&\leq \mu(E \triangle F) + \mu(F \triangle G) \\
&= \rho(E, F) + \rho(F, G)
\end{aligned}$$

with the second line following from monotonicity and the third from subadditivity.

□

**Exercise 13.** Every  $\sigma$ -finite measure is semifinite.



*Proof.* Given a  $\sigma$ -finite measure space  $(X, \mathcal{M}, \mu)$ , if  $\mu$  is finite, then semifiniteness is obvious. Assume, then, that  $\mu(X) = \infty$ , and let  $E \in \mathcal{M}$  such that  $\mu(E) = \infty$ . Also, choose  $\{E_j\}_{j \in \mathbb{N}} \subset \mathcal{M}$  such that  $\mu(E_j) < \infty$  for each  $j$  and  $\bigcup_1^\infty E_j = X$ . Then  $\bigcup_1^\infty (E \cap E_j) = E$  and by monotonicity  $\mu(E \cap E_j) \leq \mu(E_j) < \infty$  for all  $j$ . It cannot be the case that each  $E \cap E_j$  has measure 0 since

$$\infty = \mu\left(\bigcup_1^\infty (E \cap E_j)\right) \leq \sum_1^\infty \mu(E \cap E_j),$$

so we can choose  $k \in \mathbb{N}$  such that  $0 < \mu(E \cap E_k) < \infty$  as desired.  $\square$

**Exercise 14.** If  $\mu$  is a semifinite measure and  $\mu(E) = \infty$ , for any  $C > 0$  there exists  $F \subset E$  with  $C < \mu(F) < \infty$ .

[NOTE FROM CONNOR: I'm pretty proud of this proof.]

*Proof.* Let  $(X, \mathcal{M}, \mu)$  be the measure space in question. Given  $E \in \mathcal{M}$  such that  $\mu(E) = \infty$ , define  $A = \{F : F \in \mathcal{M}, F \subset E, 0 < \mu(F) < \infty\}$  and  $B = \{\mu(F) : F \in A\}$ . Now assume (contradicting the hypothesis) that  $B$  is (finitely) bounded from above. Then  $B$  has a least upper bound  $b$ .

Choose  $F_1, F_2, \dots \in A$  such that  $\{\mu(F_j)\}_{j \in \mathbb{N}}$  converges to  $b$ . This is possible because  $b$  is a limit point of  $B$ . Then define  $G_j = \bigcup_{k=1}^j F_k$  for all  $j \in \mathbb{N}$ .  $\{G_j\}$  is an ascending chain of finitely measured sets, and  $G_j \subset E$  for all  $j$ , so  $\bigcup_{j=1}^\infty G_j \subset E$  and  $\mu\left(\bigcup_{j=1}^\infty G_j\right) = \lim_{j \rightarrow \infty} \mu(G_j)$ . The sequence  $\{\mu(G_j)\}$  is monotonically increasing due to monotonicity of  $\mu$ , and it is bounded from above because  $\mu(G_j) \leq b$  for any  $j$ , so it does converge. In fact,  $\mu(F_j) \leq \mu(G_j) \leq b$  for any  $j$ , which means that  $\{\mu(G_j)\}$  is sandwiched between  $\{\mu(F_j)\}$  (which converges to  $b$ ) and the constant sequence  $\{b\}$  (which also converges to  $b$ ), so  $\{\mu(G_j)\}$  must also converge to  $b$ . Thus  $\mu\left(\bigcup_{j=1}^\infty G_j\right) = b$ .

Given this, we see that  $\mu\left(E \setminus \bigcup_{j=1}^\infty G_j\right) = \infty$ , so there is a set  $G \subset E \setminus \bigcup_{j=1}^\infty G_j$  such that  $0 < \mu(G) < \infty$ . Then  $H = G \cup \bigcup_{j=1}^\infty G_j$  is such that  $H \subset E$ ,  $0 < \mu(H) < \infty$ , and  $\mu(H) > b$ . Thus  $H \in A$  and  $\mu(H) \in B$ , which contradicts our assumption of  $b$  being a supremum of  $B$ .  $\square$

**Exercise 15.** Given a measure  $\mu$  on  $(X, \mathcal{M})$ , define  $\mu_0$  on  $\mathcal{M}$  by  $\mu_0(E) = \sup\{\mu(F) : F \subset E \text{ and } \mu(F) < \infty\}$ .

- a.  $\mu_0$  is a semifinite measure. It is called the *semifinite part* of  $\mu$ .
- b. If  $\mu$  is semifinite, then  $\mu = \mu_0$ . (Use Exercise 14.)
- c. There is a measure  $\nu$  on  $\mathcal{M}$  (in general, not unique) which assumes only the values 0 and  $\infty$  such that  $\mu = \mu_0 + \nu$ .

*Proof.*

- a. Clearly  $\mu_0(\emptyset) = 0$  since the only subset of  $\emptyset$  is  $\emptyset$ , whose measure is 0. Given a disjoint collection  $\{E_j\}_{j \in \mathbb{N}} \subset \mathcal{M}$ , we have

$$\begin{aligned} \mu_0\left(\bigcup_{j=1}^{\infty} E_j\right) &= \sup \left\{ \mu(F) : F \subset \bigcup_{j=1}^{\infty} E_j \text{ and } \mu(F) < \infty \right\} \\ &= \sup \left\{ \sum_{k=1}^{\infty} \mu(F \cap E_k) : F \subset \bigcup_{j=1}^{\infty} E_j \text{ and } \mu(F) < \infty \right\} \\ &= \sum_{k=1}^{\infty} \sup \left\{ \mu(F \cap E_k) : F \subset \bigcup_{j=1}^{\infty} E_j \text{ and } \mu(F) < \infty \right\} \end{aligned}$$

CHECK THIS SET EQUALITY CAREFULLY

$$\begin{aligned} &= \sum_{k=1}^{\infty} \sup \{ \mu(F) : F \subset E_k \text{ and } \mu(F) < \infty \} \\ &= \sum_{k=1}^{\infty} \mu_0(E_k). \end{aligned}$$

So  $\mu_0$  is a measure.

Given any  $E \in \mathcal{M}$  such that  $\mu_0(E) = \infty$ , by definition we see that

$$\infty = \sup \{ \mu(F) : F \subset E \text{ and } \mu(F) < \infty \},$$

so certainly there exists measurable  $F \subset E$  such that  $0 < \mu(F) < \infty$ . Thus  $\mu_0$  is semifinite.

- b. Note that if  $\mu(E)$  is finite, then clearly  $\mu_0(E) = \mu(E)$ , so we need only check agreement on measurable sets where  $\mu(E) = \infty$ . Given, then,  $E \in \mathcal{M}$  such that  $\mu(E) = \infty$ , by Exercise 14 we know that

$$\{ \mu(F) : F \subset E \text{ and } \mu(F) < \infty \}$$

is unbounded from above, and thus its supremum is  $\infty$ . Therefore

$$\mu_0(E) = \infty = \mu(E).$$

- c. Define  $\nu$  so that given  $E \in \mathcal{M}$ ,  $\nu(E) = 0$  if  $\mu(E) < \infty$  and  $\nu(E) = \infty$  if  $\mu(E) = \infty$ . As observed in b,  $\mu(E) = \mu_0(E)$  when  $\mu(E) < \infty$ , so

$$\mu_0(E) + \nu(E) = \mu_0(E) = \mu(E)$$

in that case. In case  $\mu(E) = \infty$ , then no matter the value of  $\mu_0(E)$ ,

$$\mu_0(E) + \nu(E) = \infty = \mu(E).$$

□

**Exercise 16.** Let  $(X, \mathcal{M}, \mu)$  be a measure space. A set  $E \subset X$  is called **locally measurable** if  $E \cap A \in \mathcal{M}$  for all  $A \in \mathcal{M}$  such that  $\mu(A) < \infty$ . Let  $\tilde{\mathcal{M}}$  be the collection of all locally measurable sets. Clearly  $\mathcal{M} \subset \tilde{\mathcal{M}}$ ; then if  $\mathcal{M} = \tilde{\mathcal{M}}$ , then  $\mu$  is called **saturated**.

- a. If  $\mu$  is  $\sigma$ -finite, then  $\mu$  is saturated.
- b.  $\tilde{\mathcal{M}}$  is a  $\sigma$ -algebra.
- c. Define  $\tilde{\mu}$  on  $\tilde{\mathcal{M}}$  by  $\tilde{\mu}(E) = \mu(E)$  if  $E \in \mathcal{M}$  and  $\tilde{\mu}(E) = \infty$  otherwise. Then  $\tilde{\mu}$  is a saturated measure on  $\tilde{\mathcal{M}}$ , called the **saturation** of  $\mu$ .
- d. If  $\mu$  is complete, so is  $\tilde{\mu}$ .
- e. Suppose that  $\mu$  is semifinite. For  $E \in \tilde{\mathcal{M}}$ , define  $\underline{\mu}(E) = \sup \{\mu(A) : A \in \mathcal{M} \text{ and } A \subset E\}$ . Then  $\underline{\mu}$  is a saturated measure on  $\tilde{\mathcal{M}}$  that extends  $\mu$ .
- f. Let  $X_1, X_2$  be disjoint uncountable sets,  $X = X_1 \cup X_2$ , and  $\mathcal{M}$  the  $\sigma$ -algebra of countable or co-countable sets in  $X$ . Let  $\mu_0$  be counting measure on  $\mathcal{P}(X_1)$ , and define  $\mu$  on  $\mathcal{M}$  by  $\mu(E) = \mu_0(E \cap X_1)$ . Then  $\mu$  is a measure on  $\mathcal{M}$ ,  $\tilde{\mathcal{M}} = \mathcal{P}(X)$ , and in the notation of parts (c) and (e),  $\tilde{\mu} \neq \underline{\mu}$ .

*Proof.*

- a. Let  $E \subset X$  be locally measurable and let  $\{E_j\}_{j \in \mathbb{N}} \subset \mathcal{M}$  such that  $\mu(E_j) < \infty$  for each  $j$  and  $\bigcup_1^\infty E_j = X$ . Then  $E = \bigcup_1^\infty (E_j \cap E)$ , and because  $E$  is locally measurable, each  $E_j \cap E$  is measurable, so  $E$  is a countable union of measurable sets. Thus  $E$  is measurable.
- b. Given  $\{E_j\}_{j \in \mathbb{N}} \subset \tilde{\mathcal{M}}$  and  $A \in \mathcal{M}$  such that  $\mu(A) < \infty$ ,

$$\left( \bigcup_1^\infty E_j \right) \cap A = \bigcup_1^\infty (E_j \cap A),$$

so  $(\bigcup_1^\infty E_j) \cap A$  is the countable union of measurable sets and is thus measurable itself.

Also, given  $E \in \tilde{\mathcal{M}}$  and  $A$  as before,

$$E^C \cap A = A \setminus (E \cap A).$$

Since  $E \cap A \in \mathcal{M}$  and  $\mathcal{M}$  is closed under differences,  $E^C \cap A \in \mathcal{M}$ .

Thus  $\tilde{\mathcal{M}}$  is a  $\sigma$ -algebra.

- c. First we show that  $\tilde{\mu}$  is a measure. Since  $\emptyset \in \mathcal{M}$ ,  $\tilde{\mu}(\emptyset) = \mu(\emptyset) = 0$ . Also, let  $\{E_j\}_{j \in \mathbb{N}} \subset \tilde{\mathcal{M}}$  be a disjoint collection. If  $\bigcup_1^\infty E_j \in \mathcal{M}$ , then since each

$E_j$  is locally measurable,  $E_k = E_k \cap \bigcup_1^\infty E_j \in \mathcal{M}$  for each  $k$ . Then

$$\begin{aligned}\tilde{\mu}\left(\bigcup_1^\infty E_j\right) &= \mu\left(\bigcup_1^\infty E_j\right) \\ &= \sum_1^\infty \mu(E_j) \\ &= \sum_1^\infty \tilde{\mu}(E_j).\end{aligned}$$

On the other hand, if  $\bigcup_1^\infty E_j \notin \mathcal{M}$ , then for some  $k \in \mathbb{N}$ ,  $E_k \notin \mathcal{M}$ . Thus

$$\begin{aligned}\tilde{\mu}\left(\bigcup_1^\infty E_j\right) &= \infty \\ &= \sum_1^\infty \tilde{\mu}(E_j)\end{aligned}$$

since  $\tilde{\mu}(E_k) = \infty$ .

Now we show that  $\tilde{\mu}$  is saturated. Let  $E \in X$  locally measurable over  $\tilde{\mu}$ . We want to show that  $E \in \tilde{\mathcal{M}}$ . Let  $A \in \mathcal{M}$  such that  $\mu(A) < \infty$ . Then since  $A \in \tilde{\mathcal{M}}$ ,  $E \cap A \in \tilde{\mathcal{M}}$ , and by monotonicity

$$\tilde{\mu}(E \cap A) < \tilde{\mu}(A) = \mu(A) < \infty.$$

Thus since  $\tilde{\mu}(E \cap A)$  is finite,  $E \cap A \in \mathcal{M}$ . Therefore  $E \in \tilde{\mathcal{M}}$ .

- d. Let  $F \subset N$  for some  $N \in \tilde{\mathcal{M}}$  where  $\tilde{\mu}(N) = 0$ . Then  $N \in \mathcal{M}$  since  $\tilde{\mu}(N)$  is finite. Since  $\mu$  is complete,  $F \in \mathcal{M} \subset \tilde{\mathcal{M}}$ . Thus  $\tilde{\mu}$  is complete.
- e. First we prove that  $\mu$  is a measure. Since  $\emptyset$  is the only subset of  $\emptyset$  and  $\mu(\emptyset) = 0$ ,  $\underline{\mu}(\emptyset) = \sup\{0\} = 0$ . Also, given a disjoint collection  $\{E_j\}_{j \in \mathbb{N}} \subset$

$\tilde{\mathcal{M}}$ ,

$$\begin{aligned}
\underline{\mu}\left(\bigcup_1^\infty E_j\right) &= \sup\left\{\mu(A) : A \in \mathcal{M} \text{ and } A \subset \bigcup_1^\infty E_j\right\} \\
&= \sup\left\{\mu\left(\bigcup_{k=1}^\infty A \cap E_k\right) : A \in \mathcal{M} \text{ and } A \subset \bigcup_1^\infty E_j\right\} \\
&= \sup\left\{\sum_{k=1}^\infty \mu(A \cap E_k) : A \in \mathcal{M} \text{ and } A \subset \bigcup_1^\infty E_j\right\} \\
&= \sum_{k=1}^\infty \sup\left\{\mu(A \cap E_k) : A \in \mathcal{M} \text{ and } A \subset \bigcup_1^\infty E_j\right\} \\
&= \sum_{k=1}^\infty \sup\{\mu(A) : A \in \mathcal{M} \text{ and } A \subset E_k\} \\
&= \sum_{k=1}^\infty \underline{\mu}(E_k).
\end{aligned}$$

Thus  $\underline{\mu}$  is a measure.

To show that it extends  $\mu$  is easy since by monotonicity, any subset of some  $E \in \mathcal{M}$  has measure less than or equal to that of  $E$ , so

$$\underline{\mu}(E) = \sup\{\mu(A) : A \in \mathcal{M} \text{ and } A \subset E\} = \mu(E).$$

Now we just need to show saturation. In particular, if  $E \in X$  is locally measurable over  $\underline{\mu}$ , then we need to show that  $E \in \tilde{\mathcal{M}}$ . Thus we take  $A \in \mathcal{M}$  such that  $\mu(A) < \infty$  and see that since  $A \in \tilde{\mathcal{M}}$ ,  $E \cap A \in \tilde{\mathcal{M}}$ . Then  $E \cap A$  is locally measurable over  $\mu$ , so  $E \cap A = (E \cap A) \cap A \in \mathcal{M}$ . Therefore  $\underline{\mu}$  is saturated.

- f. First we show that  $\mu$  is a measure on  $\mathcal{M}$ . We see that  $\mu(\emptyset) = \mu_0(\emptyset \cap X_1) = \mu_0(\emptyset) = 0$ . Let  $\{E_j\}_{j \in \mathbb{N}} \subset \mathcal{M}$  be disjoint. Then

$$\begin{aligned}
\mu\left(\bigcup_1^\infty E_j\right) &= \mu_0\left(\left(\bigcup_1^\infty E_j\right) \cap X_1\right) \\
&= \mu_0\left(\bigcup_1^\infty (E_j \cap X_1)\right) \\
&= \sum_1^\infty \mu_0(E_j \cap X_1) \\
&= \sum_1^\infty \mu(E_j).
\end{aligned}$$

Next we want to show that  $\tilde{\mathcal{M}} = \mathcal{P}(X)$ , or in other words, that every subset of  $X$  is locally measurable. Let  $E \subset X$  and  $A \in \mathcal{M}$  such that  $\mu(A) < \infty$ . Notice that  $A \cap X_1$  must be finite since  $\mu_0(A \cap X_1) < \infty$ . Thus  $A^C$  is uncountable because  $X_1$  is uncountable and  $A^C \cap X_1 = X_1 \setminus (A \cap X_1)$ . This means  $A$  cannot be co-countable, so since  $A \in \mathcal{M}$ , it must be countable. Seeing this, it is obvious that  $E \cap A$  is countable because  $E \cap A \subset A$ , so  $E \cap A \in \mathcal{M}$ .

Finally, to see that  $\tilde{\mu} \neq \underline{\mu}$ , observe that  $X_2 \notin \mathcal{M}$  because  $X_2$  is uncountable and  $X_2^C = X_1$  is uncountable. Then  $\tilde{\mu}(X_2) = \infty$  because  $X_2^2 \notin \mathcal{M}$ , but  $\mu(X_2) = \mu_0(X_2 \cap X_1) = \mu_0(\emptyset) = 0$ , so any subset of  $X_2$  also has measure 0. Thus

$$\underline{\mu}(X_2) = \sup \{ \mu(A) : A \in \mathcal{M} \text{ and } A \subset X_2 \} = \sup \{ 0 \} = 0.$$

We conclude that since  $\tilde{\mu}(X_2) = \infty \neq 0 = \underline{\mu}(X_2)$ ,  $\tilde{\mu} \neq \underline{\mu}$ .

□

**Exercise 17.** If  $\mu^*$  is an outer measure on  $X$  and  $\{A_j\}_1^\infty$  is a sequence of disjoint  $\mu^*$ -measurable sets, then  $\mu^*(E \cap (\bigcup_1^\infty A_j)) = \sum_1^\infty \mu^*(E \cap A_j)$  for any  $E \subset X$ .

*Proof.* Notice that

$$\mu^* \left( E \cap \left( \bigcup_1^\infty A_j \right) \right) = \mu^* \left( \bigcup_1^\infty E \cap A_j \right) \leq \sum_1^\infty \mu^*(E \cap A_j),$$

so to prove equality, we need only show that  $\mu^*(E \cap \bigcup_1^\infty A_j) \geq \sum_1^\infty \mu^*(E \cap A_j)$ .

Given  $n \in \mathbb{N}$ , define  $E_n = E \cap A_n$  and  $B_n = \bigcup_{j=1}^n A_j$ . Then

$$\begin{aligned} \mu^*(E \cap B_n) &= \mu^*(E \cap B_n \cap A_n) + \mu^*(E \cap B_n \cap A_n^C) \\ &= \mu^*(E \cap A_n) + \mu^*(E \cap B_{n-1}) \end{aligned}$$

when  $n > 1$ . If  $n = 1$ , then clearly  $\mu^*(E \cap B_1) = \mu^*(E \cap A_1)$  by definition. By induction, we get  $\mu^*(E \cap B_n) = \sum_{j=1}^n \mu^*(E \cap A_j)$ . Thus

$$\lim_{n \rightarrow \infty} \mu^*(E \cap B_n) = \sum_{j=1}^\infty \mu^*(E \cap A_j). \quad (1)$$

Since each  $E \cap B_n \subset E \cap \bigcup_{j=1}^\infty A_j$ , each  $\mu^*(E \cap B_n) \leq \mu^*(E \cap \bigcup_{j=1}^\infty A_j)$ , so we see that

$$\mu^*(E \cap \bigcup_{j=1}^\infty A_j) \geq \lim_{n \rightarrow \infty} \mu^*(E \cap B_n).$$

Combining with (1) yields  $\mu^*(E \cap \bigcup_{j=1}^\infty A_j) \geq \sum_{j=1}^\infty \mu^*(E \cap A_j)$ . □

**Exercise 18.** Let  $\mathcal{A} \subset \mathcal{P}(X)$  be an algebra,  $\mathcal{A}_\sigma$  the collection of countable unions of sets in  $\mathcal{A}$ , and  $\mathcal{A}_{\sigma\delta}$  the collection of countable intersections of sets in  $\mathcal{A}_\sigma$ . Let  $\mu_0$  be a premeasure on  $\mathcal{A}$  and  $\mu^*$  the induced outer measure.

- a. For any  $E \subset X$  and  $\epsilon > 0$  there exists  $A \in \mathcal{A}_\sigma$  with  $E \subset A$  and  $\mu^*(A) \leq \mu^*(E) + \epsilon$ .
- b. If  $\mu^*(E) < \infty$ , then  $E$  is  $\mu^*$ -measurable iff there exists  $B \in \mathcal{A}_{\sigma\delta}$  with  $E \subset B$  and  $\mu^*(B \setminus E) = 0$ .
- c. If  $\mu_0$  is  $\sigma$ -finite, the restriction  $\mu^*(E) < \infty$  in (b) is superfluous.

*Proof.*

- a. By the definition of  $\mu^*$ , we know there exists  $\{A_j\}_1^\infty \subset \mathcal{A}$  such that  $E \subset \bigcup_1^\infty A_j = A \in \mathcal{A}_\sigma$  where  $\sum_1^\infty \mu_0(A_j) < \mu^*(E) + \epsilon$ . Clearly  $A \subset \bigcup_1^\infty A_j$ , so by following the definition of  $\mu^*(A)$ , we get  $\mu^*(A) \leq \sum_1^\infty \mu_0(A_j)$ . Combining these two inequalities, we get

$$\mu^*(A) \leq \sum_1^\infty \mu_0(A_j) < \mu^*(E) + \epsilon$$

as desired.

- b. If  $E$  is  $\mu^*$ -measurable, we define  $B$  as follows. For each  $j \in \mathbb{N}$ , let  $B_j = \bigcup_{k=1}^\infty B_{jk}$  such that each  $B_{jk} \in \mathcal{A}$ ,  $E \subset B_j$ , and  $\sum_{k=1}^\infty \mu_0(B_{jk}) < \mu^*(E) + \frac{1}{j}$ . Let  $B = \bigcap_{j=1}^\infty B_j$ . Then by definition  $B \in \mathcal{A}_{\sigma\delta}$ , and since  $E \subset B_j$  for each  $j$ ,  $E \subset B$ . Thus  $\mu^*(E) \leq \mu^*(B)$ , and fixing  $j$  we get  $B \subset B_j$ ,  $\mu^*(E) \leq \mu^*(B) \leq \mu^*(E) + \frac{1}{j}$ . Since this is true for every  $j$ , we have  $\mu^*(E) = \mu^*(B)$ . Because  $E$  is  $\mu^*$ -measurable,

$$\begin{aligned} \mu^*(B) &= \mu^*(B \cap E) + \mu^*(B \cap E^C) \\ &= \mu^*(E) + \mu^*(B \setminus E). \end{aligned}$$

Then

$$\mu^*(E) = \mu^*(B) = \mu^*(E) + \mu^*(B \setminus E),$$

so  $\mu^*(B \setminus E) = 0$  as desired since  $\mu^*(E) < \infty$ .

Conversely, assume there exists  $B \in \mathcal{A}_{\sigma\delta}$  such that  $E \subset B$  and  $\mu^*(B \setminus E) = 0$ . Since  $B$  is a countable intersection of countable unions of elements of  $\mathcal{A}$  which are each  $\mu^*$ -measurable (Proposition 1.3),  $B$  is  $\mu^*$ -measurable. Given  $F \subset X$ , notice that

$$\begin{aligned} \mu^*(F \cap B^C) &= \mu^*(F \cap (B \cup E)^C) \\ &= \mu^*(F \cap B^C \cap E^C) \\ &= \mu^*(F \cap E^C \cap B^C) \\ &= \mu^*(F \cap E^C) - \mu^*(F \cap E^C \cap B) \\ &= \mu^*(F \cap E^C) - \mu^*(F \cap (B \setminus E)) \\ &= \mu^*(F \cap E^C). \end{aligned}$$

Then we get

$$\begin{aligned}\mu^*(F) &= \mu^*(F \cap B) + \mu^*(F \cap B^C) \\ &\geq \mu^*(F \cap E) + \mu^*(F \cap E^C),\end{aligned}$$

which shows that  $E$  is  $\mu^*$ -measurable.

- c. We need to prove the forward direction with the hypothesis that  $\mu_0$  is  $\sigma$ -finite instead of  $\mu^*(E) < \infty$ . Consider first that  $X = \bigcup_1^\infty A_j$ , a union which we can assume to be disjoint, where each  $A_j \in \mathcal{A}$  such that  $\mu_0(A_j) < \infty$ . Note that each  $A_j$  is also  $\mu^*$ -measurable since it is in  $\mathcal{A}$ .

For any  $j, n \in \mathbb{N}$ , define  $E_j = E \cap A_j$  and choose  $E_{jn} \in \mathcal{A}_\sigma$  such that  $E_j \subset E_{jn} \subset A_j$  and  $\mu^*(E_{jn}) \leq \mu^*(E_j) + \frac{1}{2^j n}$ , using part a. Note that since  $E_j$  is  $\mu^*$ -measurable,

$$\begin{aligned}\mu(E_{jn}) &= \mu^*(E_{jn} \cap E_j) + \mu^*(E_{jn} \cap E_j^C) \\ &= \mu^*(E_j) + \mu^*(E_{jn} \setminus E_j),\end{aligned}$$

so  $\mu(E_{jn}) - \mu^*(E_j) = \mu^*(E_{jn} \setminus E_j)$ . Also, by the above inequality,  $\mu^*(E_{jn} \setminus E_j) \leq \frac{1}{2^j n}$ .

Let  $B_n = \bigcup_{j=1}^\infty E_{jn}$  and  $B = \bigcap_1^\infty B_n$ . Since each  $B_n \in \mathcal{A}_\sigma$ ,  $B \in \mathcal{A}_{\sigma\delta}$ . Also, since  $E \subset B_n$  for each  $n$ ,  $E \subset B$ . Finally, we compute  $\mu^*(B \setminus E)$ :

$$\begin{aligned}\mu^*(B_n \setminus E) &= \mu^*(B_n \cap E^C) \\ &= \sum_{j=1}^\infty \mu^*(E_{jn} \cap E^C) \\ &= \sum_{j=1}^\infty \mu^*(E_{jn} \cap E_j^C) \\ &= \sum_{j=1}^\infty \mu^*(E_{jn} \setminus E_j) \\ &\leq \sum_{j=1}^\infty \frac{1}{2^j n} \\ &= \frac{1}{n}\end{aligned}$$

for all  $n$ . Thus since  $(B \setminus E) \subset (B_n \setminus E)$ ,

$$\mu^*(B \setminus E) \leq \mu^*(B_n \setminus E) \leq \frac{1}{n}.$$

This is true for all  $n$ , so  $\mu^*(B \setminus E) = 0$ .

□



**Exercise 19.** Let  $\mu^*$  be an outer measure on  $X$  induced from a finite premeasure  $\mu_0$ . If  $E \subset X$ , define the **inner measure** of  $E$  to be  $\mu_*(E) = \mu_0(X) - \mu^*(E^C)$ . Then  $E$  is  $\mu^*$ -measurable iff  $\mu^*(E) = \mu_*(E)$ . (Use Exercise 18.)

*Proof.* For the forward direction, if  $E \subset X$  is  $\mu^*$ -measurable, then

$$\begin{aligned}\mu_*(E) &= \mu_0(X) - \mu^*(E^C) \\ &= \mu^*(X) - \mu^*(E^C) \\ &= \mu^*(X \cap E) + \mu^*(X \cap E^C) - \mu^*(E^C) \\ &= \mu^*(E) + \mu^*(E^C) - \mu^*(E^C) \\ &= \mu^*(E).\end{aligned}$$

For the other direction, we start with

$$\begin{aligned}\mu^*(E) &= \mu_0(X) - \mu^*(E^C) \\ &= \mu^*(X) - \mu^*(E^C).\end{aligned}$$

Since  $\mu^*(E) = \mu^*(X \cap E) = \mu^*(X \setminus E^C)$ , we apply the above and get

$$\mu^*(X \setminus E^C) = \mu^*(X) - \mu^*(E^C).$$

We proceed to find a descending chain of sets in  $\mathcal{A}_\sigma$  whose outer measures approach that of  $E$ . By 18a, we can find for each  $n \in \mathbb{N}$  a  $B_n \in \mathcal{A}_\sigma$  such that  $E \subset B_n$  and  $\mu^*(B_n) \leq \mu^*(E) + \frac{1}{n}$ . Furthermore, we choose each  $B_n$  such that  $B_{n+1} \subset B_n$ . The existence of such a sequence is evidenced by the fact that if  $B_{n+1} \supset B_n$ , then  $B_{n+1} \cap B_n \subset B_n$ ,  $B_{n+1} \cap B_n \supset E$ , and by monotonicity,  $\mu^*(B_{n+1} \cap B_n) \leq \mu^*(B_{n+1}) \leq \mu^*(E) + \frac{1}{n}$ .

We then define  $B = \bigcap_{n=1}^{\infty} B_n$ , which is clearly in  $\mathcal{A}_{\sigma\delta}$  and contains  $E$ . It follows immediately that since  $\{B_n\}_1^{\infty}$  is a descending chain of  $\mu^*$ -measurable sets and  $\mu^*|_{\mathcal{M}}$  is a measure where  $\mathcal{M} = \{E \subset X \mid E \text{ is } \mu^*\text{-measurable}\}$ , then  $\mu^*(B) = \lim_{n \rightarrow \infty} \mu^*(B_n) = \mu^*(E)$ . To satisfy the condition from 18b, we need only prove that  $\mu^*(B \setminus E) = 0$ .

For each  $n$ , since  $B_n$  is  $\mu^*$ -measurable, we have

$$\begin{aligned}\mu^*(E^C) &= \mu^*(E^C \cap B_n) + \mu^*(E^C \cap B_n^C) \\ &= \mu^*(B_n \setminus E) + \mu^*((E \cup B_n)^C) \\ &= \mu^*(B_n \setminus E) + \mu^*(B_n^C),\end{aligned}$$

so

$$\begin{aligned}\mu^*(B_n \setminus E) &= \mu^*(E^C) - \mu^*(B_n^C) \\ &= \mu^*(X) - \mu^*(E) - \mu^*(X) + \mu^*(B_n) \\ &= \mu^*(B_n) - \mu^*(E)\end{aligned}$$

where the second equality applies the forward direction of the present theorem to  $B_n^C$ . We can now conclude that

$$\mu^*(B_n \setminus E) = \mu^*(B_n) - \mu^*(E) \leq \mu^*(E) + \frac{1}{n} - \mu^*(E) = \frac{1}{n}.$$

Since  $(B \setminus E) \subset (B_n \setminus E)$ ,  $\mu^*(B \setminus E) \leq \mu^*(B_n \setminus E) \leq \frac{1}{n}$ . This is true for all  $n$ , so  $\mu^*(B \setminus E) = 0$ , and we can use 18b to conclude that  $E$  is  $\mu^*$ -measurable.  $\square$

**Exercise 20.** Let  $\mu^*$  be an outer measure on  $X$ ,  $\mathcal{M}^*$  the  $\sigma$ -algebra of  $\mu^*$ -measurable-sets,  $\bar{\mu} = \mu^*|_{\mathcal{M}^*}$ , and  $\mu^+$  the outer measure induced by  $\bar{\mu}$  as in (1.12) (with  $\bar{\mu}$  and  $\mathcal{M}^*$  replacing  $\mu_0$  and  $\mathcal{A}$ ).

- a. If  $E \subset X$ , we have  $\mu^*(E) \leq \mu^+(E)$ , with equality iff there exists  $A \in \mathcal{M}^*$  with  $A \supset E$  and  $\mu^*(A) = \mu^*(E)$ .
- b. If  $\mu^*$  is induced from a premeasure, then  $\mu^* = \mu^+$ . (Use Exercise 18a.)
- c. If  $X = \{0, 1\}$ , there exists an outer measure  $\mu^*$  on  $X$  such that  $\mu^* \neq \mu^+$ .

*Proof.*

- a. If  $E \subset X$ , define

$$C_E = \left\{ \sum_1^\infty \mu^*(M_j) : M_j \in \mathcal{M}^* \text{ and } E \subset \bigcup_1^\infty M_j \right\}$$

so that  $\mu^+(E) = \inf C_E$  by definition. Then given  $\{M_j\}_1^\infty \subset \mathcal{M}^*$  such that  $E \subset \bigcup_1^\infty M_j$ , then

$$\mu^*(E) \leq \mu^* \left( \bigcup_1^\infty M_j \right) \leq \sum_1^\infty \mu^*(M_j).$$

Thus  $\mu^*(E)$  is a lower bound of  $C_E$ , so  $\mu^*(E) \leq \inf(C_E) = \mu^+(E)$ .

Now if there exists  $A \in \mathcal{M}^*$  such that  $A \supset E$  and  $\mu^*(A) = \mu^*(E)$ , then the collection  $\{A_j\}_1^\infty$  where  $A_1 = A$  and  $A_n = \emptyset$  for  $n > 1$  is such that  $E \subset \bigcup_1^\infty A_j$  and

$$\sum_1^\infty \mu^*(A_j) = \mu^*(A) = \mu^*(E).$$

Thus  $\mu^*(E) \in C_E$  and  $\mu^*(E)$  is a lower bound of  $C_E$ , so  $\mu^*(E) = \inf C_E = \mu^+(E)$ .

On the other hand, if  $\mu^*(E) = \mu^+(E)$ , then choose for  $n \in \mathbb{N}$  an  $A_n \in \mathcal{M}^*$  such that  $E \subset A_n$  and  $\mu^*(A_n) \leq \mu^+(E) + \frac{1}{n} = \mu^*(E) + \frac{1}{n}$  (Exercise 18a). Then clearly  $\mu^*(\bigcap_1^\infty A_n) = \mu^*(E)$ , and  $\bigcap_1^\infty A_n \in \mathcal{M}^*$  since  $\mathcal{M}^*$  is a  $\sigma$ -algebra.

- b. Let  $\mu_0 : \mathcal{A} \rightarrow [0, \infty]$  be the premeasure which induces  $\mu^*$ . Given  $E \subset X$ , by Exercise 18a we can choose for any  $n \in \mathbb{N}$  an  $A_n \in \mathcal{A}_\sigma$  such that  $E \subset A_n$  and  $\mu^*(A_n) \leq \mu^*(E) + \frac{1}{n}$ . Let  $A = \bigcap_1^\infty A_n$ . Then  $\mu^*(A) \leq \mu^*(A_n) \leq \mu^*(E) + \frac{1}{n}$  for all  $n$ , so  $\mu^*(A) = \mu^*(E)$ . Since any element of  $\mathcal{A}$  is  $\mu^*$ -measurable and  $A$  is a countable intersection of countable unions of elements of  $\mathcal{A}$ ,  $A \in \mathcal{M}^*$ . Thus by part a of the present exercise,  $\mu^*(E) = \mu^+(E)$ , and since  $E$  varies over  $\mathcal{P}(X)$ ,  $\mu^* = \mu^+$ .
- c. Consider the outer measure  $\mu^*$  defined by

$$\begin{aligned}\emptyset &\mapsto 0 \\ \{0\} &\mapsto 2 \\ \{1\} &\mapsto 2 \\ X &\mapsto 3.\end{aligned}$$

The proof that this is an outer measure is routine. Notice that  $\{0\}$  is not  $\mu^*$ -measurable since

$$\mu^*(X \cap \{0\}) + \mu^*(X \cap \{1\}) = 4 > 3 = \mu^*(X).$$

Thus  $\{1\}$  is also not  $\mu^*$ -measurable, so  $\mathcal{M}^* = \{X, \emptyset\}$ . Since  $X$  is the only member of  $\mathcal{M}$  containing  $\{0\}$  or  $\{1\}$ , we can see that  $\mu^+$  is described by

$$\begin{aligned}\emptyset &\mapsto 0 \\ \{0\} &\mapsto 3 \\ \{1\} &\mapsto 3 \\ X &\mapsto 3.\end{aligned}$$

Thus  $\mu^* \neq \mu^+$ .

□

**Exercise 21.** Let  $\mu^*$  be an outer measure induced from a premeasure and  $\bar{\mu}$  the restriction of  $\mu^*$  to the  $\mu^*$ -measurable sets. Then  $\bar{\mu}$  is saturated. (Use Exercise 18.)

*Proof.* Let  $\mu_0 : \mathcal{A} \rightarrow [0, \infty]$  be the premeasure from which  $\mu^*$  is induced. Given  $E \subset X$  locally measurable (note that  $E^C$  is also locally measurable by 16b), we want to show that  $E$  is  $\mu^*$ -measurable. We thus consider  $F \subset X$  such that  $\mu^*(F) < \infty$  and wish to show that  $\mu^*(F) \geq \mu^*(F \cap E) + \mu^*(F \cap E^C)$ .

By 18a, given  $n \in \mathbb{N}$ , we can find  $B_n \in \mathcal{A}_\sigma$  such that  $F \subset B_n$  and  $\mu^*(B_n) \leq \mu^*(F) + \frac{1}{n}$ . Fix  $n \in \mathbb{N}$ . Then  $B_n$  is  $\mu^*$ -measurable, so we can use the finite additivity of  $\bar{\mu}$  to get

$$\begin{aligned}\mu^*(B_n) &= \bar{\mu}(B_n) \\ &= \bar{\mu}(B_n \cap E) + \bar{\mu}(B_n \cap E^C) \\ &= \mu^*(B_n \cap E) + \mu^*(B_n \cap E^C) \\ &\leq \mu^*(F) + \frac{1}{n}.\end{aligned}$$

Since  $(F \cap E) \subset (B_n \cap E)$  and  $(F \cap E^C) \subset (B_n \cap E^C)$ , monotonicity together with the final inequality from above shows us that

$$\begin{aligned}\mu^*(F \cap E) + \mu^*(F \cap E^C) &\leq \mu^*(B_n \cap E) + \mu^*(B_n \cap E^C) \\ &\leq \mu^*(F) + \frac{1}{n}.\end{aligned}$$

Letting  $n$  vary over  $\mathbb{N}$ , we conclude that

$$\mu^*(F \cap E) + \mu^*(F \cap E^C) \leq \mu^*(F).$$

Thus  $E$  is  $\mu^*$ -measurable, so  $\bar{\mu}$  is saturated.  $\square$

**Exercise 22.** Let  $(X, \mathcal{M}, \mu)$  be a measure space,  $\mu^*$  the outer measure induced by  $\mu$  according to (1.12),  $\mathcal{M}^*$  the  $\sigma$ -algebra of  $\mu^*$ -measurable sets, and  $\bar{\mu} = \mu^*|_{\mathcal{M}^*}$ .

- a. If  $\mu$  is  $\sigma$ -finite, then  $\bar{\mu}$  is the completion of  $\mu$ . (Use Exercise 18.)
- b. In general,  $\bar{\mu}$  is the saturation of the completion of  $\mu$ . (See Exercises 16 and 21.)

*Proof.*

- a. Let  $\mathcal{N} = \{N \in \mathcal{M} : \mu(N) = 0\}$  and

$$\overline{\mathcal{M}} = \{E \cup F : E \in \mathcal{M}, F \subset X \text{ such that } F \subset N \text{ for some } N \in \mathcal{N}\}.$$

We need to show that  $\bar{\mu}$  is the unique complete measure extending  $\mu$  on  $\overline{\mathcal{M}}$ . Thus we need to prove two things, the first of which is that  $\mathcal{M}^* = \overline{\mathcal{M}}$ , and the second of which is that  $\bar{\mu}$  extends  $\mu$ .

First, assume  $M \in \mathcal{M}^*$ , so  $M$  is  $\mu^*$ -measurable. Then by 18c, we can find  $B \in \mathcal{M}$  ( $\mathcal{M} = \mathcal{M}_{\sigma\delta}$  since  $\mathcal{M}$  is a  $\sigma$ -algebra) such that  $M \subset B$  and  $\mu^*(B \setminus M) = 0$ . Since both  $B$  and  $M$  are  $\mu^*$ -measurable,  $B \setminus M$  is also  $\mu^*$ -measurable, so we can repeat the process on  $B \setminus M$  by finding  $C \in \mathcal{M}$  such that  $(B \setminus M) \subset C$  and  $\mu^*(C \setminus (B \setminus M)) = 0$ . Notice that  $C \cap M$  is  $\mu^*$ -measurable and

$$\begin{aligned}0 &= \mu^*(C \setminus (B \setminus M)) \\ &= \mu^*(C \cap (B \cap M^C)^C) \\ &= \mu^*(C \cap (B^C \cup M)) \\ &= \mu^*((C \cap B^C) \cup (C \cap M)),\end{aligned}$$

so by monotonicity,  $\mu^*(C \cap M) = 0$ . Thus we apply 18c one more time to get a  $D \in \mathcal{M}$  such that  $(C \cap M) \subset D$  and  $\mu^*(D \setminus (C \cap M)) = 0$ . Then

$$\mu^*(D) \leq \mu^*(D \setminus (C \cap M)) + \mu^*(C \cap M) = 0,$$

so in fact  $D \in \mathcal{N}$ , making  $C \cap M$  a subset of an element of  $\mathcal{N}$ . Now all we need to know is that  $(B \setminus C) \cup (C \cap M) = M$ , and we will have shown that  $M \in \overline{\mathcal{M}}$ . The forward inclusion is easy, since if  $x \in B \setminus C$  then  $x \in M$  because  $C$  contains all the elements of  $B$  not in  $M$ , and clearly  $C \cap M \subset M$ . For the other direction, if  $x \in M$ , then either  $x \in C$  and thus  $x \in C \cap M$ , or  $x \notin C$ , in which case since  $x \in M \subset B$ ,  $x \in B \setminus C$ .

Now suppose  $M \in \overline{\mathcal{M}}$ , so  $M = E \cup F$  for some  $E \in \mathcal{M}$  and  $F \in X$  such that  $F \subset N$  for some  $N \in \mathcal{N}$ . Given  $G \subset X$  such that  $\mu^*(G) < \infty$ , we want to show that

$$\mu^*(G) \geq \mu^*(G \cap M) + \mu^*(G \cap M^C).$$

This is straightforward:

$$\begin{aligned} \mu^*(G \cap M) + \mu^*(G \cap M^C) &= \mu^*(G \cap (E \cup F)) + \mu^*(G \cap (E \cup F)^C) \\ &= \mu^*((G \cap E) \cup (G \cap F)) + \mu^*(G \cap E^C \cap F^C) \\ &\leq \mu^*(G \cap E) + \mu^*(G \cap F) + \mu^*(G \cap E^C) \\ &= \mu^*(G \cap E) + \mu^*(G \cap E^C) \\ &= \mu^*(G) \end{aligned}$$

where the final equation follows because  $E$  is  $\mu^*$ -measurable.

Finally, we need  $\overline{\mu}(M) = \mu(M)$  for  $M \in \mathcal{M}$ . This follows immediately from (1.13), taking  $\mu$  to be  $\mu_0$  and  $\mathcal{M}$  to be  $\mathcal{A}$ .

- b.** Let  $\overline{\mathcal{M}}$ ,  $\mathcal{M}^*$ , and  $\mathcal{N}$  be defined as above, and  $\widetilde{\mathcal{M}}$  the set of all locally measurable sets under  $\overline{\mu}$  where  $\overline{\mu}$  is the completion of  $\mu$ . Then we need to show that  $\mathcal{M}^* = \widetilde{\mathcal{M}}$  and that  $\overline{\mu}(E) = \mu(E)$  if  $E \in \mathcal{M}$  and  $\overline{\mu}(E) = \infty$  otherwise. Note that the proof of part **a** only used  $\sigma$ -finiteness to show that  $\mathcal{M}^* \subset \overline{\mathcal{M}}$ , so we can use the facts that  $\overline{\mathcal{M}} \subset \mathcal{M}^*$  and  $\mu^*(M) = \overline{\mu}(M)$  for all  $M \in \mathcal{M}^*$ .

First, let  $M \in \mathcal{M}^*$ . Then given  $E \in \overline{\mathcal{M}}$ ,  $\mu^*(E) = \mu^*(E \cap M) + \mu^*(E \cap M^C)$

Now let  $M \in \widetilde{\mathcal{M}}$ . Given  $E \subset X$  such that  $\mu^*(E) < \infty$  and  $F \in \overline{\mathcal{M}}$ ,

$$\begin{aligned} \mu^*(E \cap M) + \mu^*(E \cap M^*) &= \dots \\ &= \dots \\ &\leq \dots \\ &= \dots \\ &= \mu^*(E) \end{aligned}$$

$$\begin{aligned} \mu^*(E) &= \mu^*(E \cap F \cap M) + \mu^*(E \cap (F \cap M)^C) \\ &= \mu^*(E \cap F \cap M) + \mu^*(E \cap (F^C \cup M^C)) \\ &= \mu^*(E \cap F \cap M) + \mu^*((E \cap F^C) \cup (E \cap M^C)) \\ &\leq \mu^*(E \cap F \cap M) + \mu^*(E \cap F^C) + \mu^*(E \cap M^C) \end{aligned}$$

UNFINISHED

□

**Exercise 25.** Complete the proof of Theorem 1.19.

*Proof.* For each  $j \in \mathbb{Z}$ , let  $E_j = E \cap (j, j+1]$ . Then  $\bigcup_1^\infty E_j = E$  and we can find open  $U_{jk}$  for each  $j, k \in \mathbb{Z}$  such that  $(\bigcap_{k=1}^\infty U_{jk}) \setminus N_j = E_j$  for some null set  $N_j$ . Then each  $\bigcup_{j \in \mathbb{Z}} U_{jk}$  is open and contains  $E$ . Define  $V = \bigcap_{k=1}^\infty \bigcup_{j \in \mathbb{Z}} U_{jk}$ . Then  $V$  is a  $G_\delta$  set containing  $E$ , and  $V \setminus (V \setminus E) = E$ , so we simply need to show that  $\mu(V \setminus E) = 0$ . Observe that

$$\begin{aligned} V \setminus E &= V \cap \left( \bigcup_{j \in \mathbb{Z}} E_j \right)^C \\ &= V \cap \left( \bigcap_{j \in \mathbb{Z}} E_j^C \right) \\ &= \left( \bigcap_{j \in \mathbb{Z}} V \cap E_j^C \right) \\ &= \left( \bigcap_{j \in \mathbb{Z}} V \setminus E_j \right) \\ &\subset V \setminus E_1 \\ &\subset N_1, \end{aligned}$$

so by monotonicity  $\mu(V \setminus E) = 0$ .

For  $(a \implies c)$ , define  $E_j$  as before and let  $H_{jk}$  be closed for each  $j, k \in \mathbb{Z}$  such that each  $E_j = (\bigcup_{k=1}^\infty H_{jk}) \cup N_j$  for some null  $N_j$ . Let  $H = \bigcup_{j \in \mathbb{Z}} \bigcup_{k=1}^\infty H_{jk}$  and  $N = \bigcup_{j \in \mathbb{Z}} N_j$ . Then  $H$  is an  $F_\sigma$  set and  $H \subset E$ . Also,  $\mu(N) = 0$  by subadditivity. Observe, then, that

$$\begin{aligned} E &= \bigcup_{j \in \mathbb{Z}} E_j \\ &= \bigcup_{j \in \mathbb{Z}} \bigcup_{k=1}^\infty H_{jk} \cup N_j \\ &= \left( \bigcup_{j \in \mathbb{Z}} \bigcup_{k=1}^\infty H_{jk} \right) \cup \bigcup_{j \in \mathbb{Z}} N_j \\ &= H \cup N. \end{aligned}$$

Then it becomes obvious that  $E \setminus H \subset N$ , so  $\mu(E \setminus H) = 0$ . □

**Exercise 26.** Prove Proposition 1.20. (Use Theorem 1.18.)

*Proof.* Given  $\epsilon > 0$ , choose  $U \supset E$  be open such that  $\mu(U) < \mu(E) + \epsilon/2$  and  $H \subset E$  compact such that  $\mu(H) > \mu(E) - \epsilon/2$ . Then express  $U$  as a countable union of open intervals  $\bigcup_1^\infty U_j = U$ . Since  $H \subset E \subset U$ ,  $\{U_j\}_1^\infty$  is an open cover of  $H$ , so there is a finite subcover  $\{U_j\}_1^n$  for some  $n$  after potential reordering. Define  $A = \bigcup_1^n U_j$  and observe that  $H \subset A \subset U$ . Then

$$\mu(E) - \epsilon/2 < \mu(H) \leq \mu(A) \leq \mu(U) < \mu(E) + \epsilon/2.$$

We now observe that

$$E \cup A = (E \triangle A) \cup (E \cap A),$$

so

$$\mu(E \triangle A) = \mu(E \cup A) - \mu(E \cap A).$$

Clearly  $H \subset (E \cup A) \subset U$  and  $H \subset (E \cap A) \subset U$ , so

$$\begin{aligned} \mu(E \triangle A) &= \mu(E \cup A) - \mu(E \cap A) \\ &< \mu(E) + \epsilon/2 - (\mu(E) - \epsilon/2) \\ &= \epsilon \end{aligned}$$

as desired.  $\square$

**Exercise 27.** Prove Proposition 1.22a. (Show that if  $x, y \in C$  and  $x < y$ , there exists  $z \notin C$  such that  $x < z < y$ .)

*Proof.* SKIP FOR NOW  $\square$

**Exercise 28.** Let  $F$  be increasing and right continuous, and let  $\mu_F$  be the associated measure. Then  $\mu_F(\{a\}) = F(a) - F(a-)$ ,  $\mu_F([a, b]) = F(b) - F(a-)$ ,  $\mu_F((a, b]) = F(b) - F(a)$ , and  $\mu_F((a, b)) = F(b-) - F(a)$ .

*Proof.* First, consider an increasing sequence  $\{a_n\}_1^\infty$  such that each  $a_n < a$  and  $\lim_{n \rightarrow \infty} a_n = a$ . Then

$$\begin{aligned} \mu_F(\{a\}) &= \mu_F\left(\bigcap_1^\infty (a_n, a]\right) \\ &= \lim_{n \rightarrow \infty} \mu_F((a_n, a]) \\ &= \lim_{n \rightarrow \infty} F(a) - F(a_n) \\ &= F(a) - F(a-). \end{aligned}$$

Next we consider  $(a, b)$ . Let  $\{b_n\}_1^\infty$  be an increasing sequence whose limit is  $b$  where each  $b_n < b$ . Then

$$\begin{aligned} \mu_F((a, b)) &= \mu_F\left(\bigcup_1^\infty (a, b_n]\right) \\ &= \lim_{n \rightarrow \infty} \mu_F(a, b_n] \\ &= \lim_{n \rightarrow \infty} F(b_n) - F(a) \\ &= F(b-) - F(a). \end{aligned}$$

The other two equalities follow immediately from these two, since

$$\begin{aligned}
\mu_F([a, b)) &= \mu_F(\{a\} \cup (a, b)) \\
&= \mu_F(\{a\}) + \mu_F((a, b)) \\
&= F(a) - F(a-) + F(b-) - F(a) \\
&= F(b-) - F(a-),
\end{aligned}$$

and by similar logic,

$$\begin{aligned}
\mu_F([a, b]) &= \mu_F([a, b)) + \mu_F(\{b\}) \\
&= F(b-) - F(a-) + F(b) - F(b-) \\
&= F(b) - F(a-).
\end{aligned}$$

□

**Exercise 29.** Let  $E$  be a Lebesgue measurable set.

- a. If  $E \subset N$  where  $N$  is the nonmeasurable set described in §1.1, then  $m(E) = 0$ .
- b. If  $m(E) > 0$ , then  $E$  contains a nonmeasurable set. (It suffices to assume  $E \subset [0, 1]$ . In the notation of §1.1,  $E = \bigcup_{r \in R} E \cap N_r$ .)

*Proof.*

- a. Given  $r \in R$ , define  $E_r$  to be

$$E_r = \{x + r : x \in E \cap [0, 1 - r)\} \cup \{x + r - 1 : x \in E \cap [1 - r, 1)\}.$$

Immediately, we can see that

$$\begin{aligned}
m(E_r) &= m(\{x + r : x \in E \cap [0, 1 - r)\} \cup \{x + r - 1 : x \in E \cap [1 - r, 1)\}) \\
&= m(\{x + r : x \in E \cap [0, 1 - r)\}) + m(\{x + r - 1 : x \in E \cap [1 - r, 1)\}) \\
&= m(\{x : x \in E \cap [0, 1 - r)\}) + m(\{x : x \in E \cap [1 - r, 1)\}) \\
&= m(\{x : x \in E \cap [0, 1 - r)\} \cup \{x : x \in E \cap [1 - r, 1)\}) \\
&= m(E),
\end{aligned}$$

where the third equality comes by the translation invariance of  $m$ . Using a similar argument to that of §1.1, we see that

$$m\left(\bigcup_{r \in R} E_r\right) \leq m([0, 1)) = 1,$$

but

$$m\left(\bigcup_{r \in R} E_r\right) = \sum_{r \in R} m(E_r) = \sum_{r \in R} m(E).$$

Clearly, the final expression above cannot be finite unless  $m(E) = 0$ , so since it is forced to be  $\leq 1$  (i.e. finite), in fact  $m(E) = 0$ .



**b.** Since  $X = \bigcup_{z \in \mathbb{Z}} I_z$  where each  $I_z = [z, z + 1)$ ,

$$0 < m(E) = m\left(\bigcup_{z \in \mathbb{Z}} I_z \cap E\right) = \sum_{z \in \mathbb{Z}} m(I_z \cap E),$$

so  $m(I_j \cap E) > 0$  for some  $j \in \mathbb{Z}$ . Then

$$(I_j \cap E) - j \subset I_0$$

and

$$m(I_j \cap E) - j = m(I_j \cap E) > 0,$$

so since  $I_0 = \bigcup_{r \in R} N_r$ , there exists a  $k$  such that

$$m(((I_j \cap E) - j) \cap N_k) > 0$$

Denote  $((I_j \cap E) - j) \cap N_k$  by  $F$ . Note that part **a.** holds for each  $N_r$  in §1.1 since the choice of  $N$  was arbitrary and each  $N_r$  consists of exactly one representative from each equivalence class. It follows from **a.**, then, that since  $F \subset N_k$  and  $m(F) > 0$ ,  $F$  is not Lebesgue measurable.

Finally, notice that  $F + j \subset I_j \cap E \subset E$ , so  $F + j$  is a nonmeasurable set contained in  $E$  since if  $F + j$  were measurable,  $F$  would be as well.

□

**Exercise 30.** If  $E \in \mathcal{L}$  and  $m(E) > 0$ , for any  $\alpha < 1$  there is an open interval  $I$  such that  $m(E \cap I) > \alpha m(I)$ .

*Proof.* UNFINISHED

□